

Work Breakdown Structure (WBS)

Municipal Solid Waste Route Optimizer

Hierarchical Work Package Decomposition & Schedule | Student SRN: PES1UG24CS453

1. Executive Summary & Project Hierarchy

The Work Breakdown Structure (WBS) decomposes the development and deployment of the Municipal Solid Waste Route Optimizer into discrete, manageable deliverables and work packages across 8 key phases. Each phase defines specific milestones, deliverables, and role ownership to guarantee adherence to the sub-45s optimization SLA and municipal deployment standards.

2. Hierarchical Work Breakdown Structure (WBS Tree)

1.0 Project Management & Quality Assurance

- 1.1 Project Initiation, Scope Definition & PES SE Lab Charter
- 1.2 Agile Sprint Cadence Planning (2-week iterations, daily standups)
- 1.3 Risk Management, Quality Gates & Compliance Audits

2.0 Requirements Engineering & Analysis

- 2.1 Stakeholder Elicitation (Sanitation Supervisors, Drivers, Municipal Admin)
- 2.2 Formal IEEE 830 Software Requirements Specification (SRS)
- 2.3 Use Case Modeling, Actor Hierarchy & Requirement Traceability Matrix

3.0 System Architecture & Microservices Design

- 3.1 Architectural Pattern Evaluation & Microservices Selection
- 3.2 UML 2.0 Component Modeling (Interfaces, Ball & Socket, Connectors)
- 3.3 Polyglot Persistence Architecture (PostGIS, TimescaleDB, Redis Cache)
- 3.4 API Gateway, Service Mesh & Edge Security Specification

4.0 IoT Telemetry Ingestion & Sensor Integration

- 4.1 Ultrasonic Sensor Calibration & Payload Definition (MQTT/NB-IoT)
- 4.2 Sensor Data Ingestion Service & Telemetry Pipeline
- 4.3 Dynamic Threshold Filtering Engine (Isolate Bins >80% Full)
- 4.4 Telemetry Time-Series Persistence & Anomaly Noise Removal

5.0 Route Optimization & Algorithmic Engine

- 5.1 Urban Road Network Topology & Graph Formulation
- 5.2 Multi-Vehicle Routing Problem (VRP) Heuristic Solver
- 5.3 Distributed GPU/Compute Acceleration (Sub-45s / 50 Trucks / 10k Bins)
- 5.4 Route Request Assembly Interface (gRPC / Protobuf Endpoint)

6.0 Manifest Dispatch & User Interface Portals

- 6.1 Supervisor Web Operations Portal (GIS Fleet Tracking & Analytics)
- 6.2 Driver In-Cab Navigation Mobile App (WebSocket Manifest Sync)
- 6.3 Citizen Missed-Pickup Reporting Portal & Ticket Tracking
- 6.4 Notification & Incident Alerting Service

7.0 Verification, Validation & Testing

- 7.1 Automated Unit & Component Integration Testing
- 7.2 Rigorous 45-Second VRP Performance & Stress Benchmark Testing
- 7.3 Security Penetration Testing, JWT Validation & Role-Based Access Audit

- 7.4 Field Acceptance Testing with Sanitation Drivers & Hardware Sensors

8.0 Deployment, DevOps & Infrastructure

- 8.1 Containerization with Docker & Multi-Stage Production Builds
- 8.2 Kubernetes Cluster Orchestration & Horizontal Pod Autoscaling (HPA)
- 8.3 CI/CD Deployment Automation & GitOps Pipelines
- 8.4 Centralized Observability, Logging & Alerting (Prometheus / Grafana)

3. Work Package Dictionary & Milestone Matrix

WBS ID	Work Package	Key Deliverable	Owner Role	Duration
WP 1.0	Project Management	Agile Plan, Risk Matrix, Sprint Logs	Project Lead	2 Weeks
WP 2.0	Requirements Eng.	IEEE 830 SRS, Traceability Matrix	Systems Analyst	2 Weeks
WP 3.0	System Architecture	UML Component Diagram, Arch Justification	Chief Architect	2 Weeks
WP 4.0	IoT Telemetry Ingest	Ingestion Service, MQTT Adapter, Filter	IoT / Backend Lead	3 Weeks
WP 5.0	Route Optimization	VRP Solver, <45s Benchmark Engine, gRPC	Algorithm Specialist	4 Weeks
WP 6.0	Portals & Manifest	Supervisor Dashboard, Driver App, WS Stream	Frontend / Mobile	3 Weeks
WP 7.0	Testing & QA	Test Plan, 10k Bin Load Tests, Bug Tracker	QA / Security Lead	2 Weeks
WP 8.0	DevOps & Launch	K8s Manifests, CI/CD Pipeline, Grafana	DevOps Engineer	2 Weeks